

AMENDMENT NO. 1 SEPTEMBER 2011
TO
IS 7079 : 2008 AUTOMOTIVE VEHICLES —
BRAKE HOSE ASSEMBLIES FOR HYDRAULIC BRAKING
SYSTEMS USED WITH NON-PETROLEUM BASE
BRAKE FLUID — SPECIFICATION

(Third Revision)

(Second cover page, Foreword, para 2) — Add following para after para 2:

‘For extending approval as per IS 7079 : 1995 to approval as per IS 7079 : 2008, only the following verification/tests need be carried out:

- a) Expansion test at 20 MPa pressure as per **5.5**,
- b) Static Ozone Resistance Test as per **5.13**,
- c) Dynamic Ozone Resistance Test as per **5.14**,
- d) Hot Impulse Strength Test as per **5.15**,
- e) Brake Fluid Compatibility Test as per **5.16**, and
- f) HL or HR marking as per **7.1**.’

(Second cover page, Foreword, para 5) — Substitute ‘Annex K’ for ‘Annex J’

(Page 1, clause 1.1) — Substitute the following for the existing:

‘1.1 This standard specifies the performance requirements, test procedures and methods of sampling of brake hose assemblies to be used in hydraulic braking systems of automotive vehicles having nominal internal diameter upto 5 mm.’

(Page 1, clause 3.11) — Add following new clause after 3.11:

‘3.12 Type of Brake Hose Assembly

Brake hose assemblies which do not differ in the details of following parameters are considered as same type of brake hose assembly:

- i) Brake hose construction,
- ii) Material, and
- iii) Fitting crimp or swage area.

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Changes in parameters, which do not influence the integrity of the hose coupling joints, such as the following do not constitute a different type:

- i) Length;
- ii) End fitting shape, thread size, port dimensions, hexagon size; and
- iii) External appendage such as protective sleeve, armour spring, centre fitting.'

(Page 2, clauses 4.1.1 and 4.1.2) — Substitute the following for the existing:

4.1.1 Lining

The lining shall consist of a suitable material (such as rubber compound, Teflon etc.) resistant to non-mineral oil based brake fluid.

4.1.2 Reinforcement

The reinforcement shall consist of suitable material such as textile (cotton, viscose rayon, polyester fibre or any other suitable synthetic fibre or a combination of these) or metallic wire or combination thereof etc.'

(Page 2, clause 4.2.1) — Insert the following at the end of clause:

'These conditions are deemed to be satisfied, if the requirements of 5.4 are complied with.'

(Page 2, clause 4.2.2, second sentence) — Substitute 'There shall be at least two reinforcement plies and the same shall be impregnated with a suitable compound.' for 'There shall be at least two reinforcement plies and the same shall be impregnated with a suitable rubber compound.'

(Page 2, clause 5.1, last para) — Substitute the following for the existing:

'Further, in case the hose length is more than what required for tests, shorter hose length shall be prepared for such tests. Hose assembly configuration is such that it is impractical to conduct any of the tests, hose assemblies produced with straight end fittings shall be used as substitute brake hose assemblies for such tests.'

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(Page 2, clause 5.2.1.1) — Substitute the following for the existing:

‘5.2.1.1 Criteria for approval

Fifteen samples shall be submitted for testing along with the relevant data. These samples shall be tested in the sequence given in Table 1.

The test schedule shall consist of all the tests listed in Table 1. The complete test schedule shall be conducted for each type of brake hose assembly.’

[Page 3, Table 1, Sl No. (ii)] — Substitute ‘Constriction’ for ‘Construction’

(Page 3, Table 1) — Add the following Note at end of table:

‘NOTE — Tests on samples 3, 4 and 5 (Bursting strength, whip test and tensile tests) need not be carried out if the condition prescribed in 5.7, 5.8 and 5.9 are satisfied.’

(Page 4, clause 5.7) — Delete the NOTE given below 5.7.

(Page 4, clause 5.7) — Add the following at the end of the clause:

‘This test need not be carried out if it has been carried out as part of tests described in 5.10 or 5.16.

(Page 4, clauses 5.8 and 5.9) — Add the following at the end of both the clauses:

‘This test need not be carried out if it has been carried out as part of test described in 5.10.’

(Page 4, clause 5.10) — Add the following Note at the end of the clause:

‘NOTE — This test is applicable only when the material for hose is a compound of rubber.’

(Page 5, clause 6.1.1) — Substitute the following for the existing:

‘6.1.1 Lot

In any consignment, the lengths of assembled brake hose needed for carrying out inspection or tests, of the same type, size and diameter, having the same number of plies, produced under essentially similar conditions of manufacture (such as

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those from single batch of raw material or from components obtained from a single source or from a single production method or undergoing a single curing process) shall be separated into groups and each such group shall constitute a separate lot.’

(Page 6, clause 7.1, first sentence) — Substitute the following for the existing:

‘Each brake hose assembly (*see* 3.1) shall be legibly and indelibly marked with the following:’

(Page 6, clause 7.2) — Insert following new clauses after 7.2:

‘8 TYPE APPROVAL FOR STATUTORY CERTIFICATION

If the tests are carried out for the purpose of statutory certification, the test authority shall issue a type approval certificate if the brake hoses are found to comply with the requirements of test given in 5.2.1.

8.1 Technical Information to be Submitted by the Applicant

The applicant shall submit the technical specification of the brake hose assembly contained in Annex J to the test agency for the purpose of statutory certification.

8.1.1 Every modification pertaining to the information, even if the changes are not technical in nature declared in accordance with 8.1 shall be intimated by the manufacturer to the certifying agency.

If the changes in parameters are not related to the provisions, no further action need be taken.

If the changes in parameters are related to the tests, the Test Agency, which has issued the certificate of compliance, shall then consider, whether:

- i) the brake hose assembly with the changed specifications still complies with provisions,
- or
- ii) Any further verification is required to establish compliance.

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8.2 Criteria for Extension of Approval

For considering whether testing is required or not, guidelines given in Table 7 shall be used. Changes other than those listed in Table 7 shall not be considered to affect the performance of brake hose adversely.

Table 7 Criteria for Extension of Approval
(Clause 8.2)

Sl No.	Parameter and Change	Tests to be Conducted
1.	Change of material of hose or type of braiding	Burst test and whip test to be conducted
2.	Change in the crimping or swaging area	Whip test to be conducted
3.	Section dimensions of the hose	Burst test and whip test to be conducted
4.	Change in bore diameter	All tests except salt spray test
5.	End fittings	Salt spray test need not be carried out if the manufacturer can show evidence that the corrosion protection is same. In case the braiding material is changed to metallic other than stainless steel, salt spray test shall be carried out.

NOTE — Changes other than above shall not call for re-testing of the Brake Hose Assembly.

8.2.1 In case of fulfillment of criterion **8.1.1(i)** or after results of further verification as per **8.1.1(ii)** are satisfactory, the approval of compliance shall be extended for the changes carried out.

8.3 Conformity of Production (COP)

The frequency for carrying out check/tests by test agency shall be once in two years. Tests applicable for the purpose of COP, shall be those listed in **5.2.3.**

(Page 6, clause 8) — Renumber the existing clause as **9**.

(Page 13, Annex H) — Add the following new Annex J at the end:

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‘ANNEX J

**INFORMATION TO BE SUBMITTED AT THE TIME OF
APPLICATION FOR APPROVAL, IF REQUIRED FOR STATUTORY
PURPOSES
(Clause 8.1)**

J-1	Trade name or mark of the brake hose assembly:
J-2	Manufacturer's name and address:
J-3	Telephone No:
J-4	FAX. No:
J-5	E mail address:
J-6	Contact person:
J-7	If applicable, name and address of manufacturer's representative:
J-8	Nominal internal diameter:
J-9	Length:
J-10	Material of lining (for example, rubber compound, Teflon etc):
J-11	The material of reinforcement (Textile/metal wire/comboination thereof):
J-11.1	If textile, type of material (cotton/viscose rayon/polyester fibre/ other synthetic fibre):
J-11.2	If metal, wire material (stainless steel/other):
J-12	Material of end connection:
J-13	Surface treatment for the end connection:
J-14.0	Crimping details:
J-14.1	Method:
J-14.2	Length:
J-14.3	Diameter:
J-14.4	Location of Markings on brake hose assembly:
NOTE — Information regarding J-8 to J-14.4 may be given in a drawing.	

(Page 15, Annex J) — Substitute ‘Annex K’ for ‘Annex J’.

(TED 4)